Zelin LI

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Education

2020 – 2026 **Ph.D. in Electrical Engineering, City University of Hong Kong, HK.**
Thesis title: *Intelligent Processing and Analysis of Caenorhabditis elegans Microscopic Images.*

2016 – 2020 **B.Eng. in EIE (Major), Huazhong University of Science and Technology, CN**
Thesis title: *Blockchain Architecture Design for Smart Healthcare Scenarios.*

2017 – 2020 **B.S. in Finance (Minor), Huazhong University of Science and Technology, CN**
Thesis title: *A Study on the Impact of FinTech on the Stock Performance of Chinese Commercial Banks: A Text-Based Analysis of Annual Reports.*

Positions and Experiences

2025 – Present **Postdoctoral Researcher** SIGS, Tsinghua University, CN.

2025 – 2025 **Research Associate** City University of Hong Kong, HK.

Research Associate CityUHK & Oxford Joint Research Center (CIMDA), HK.

2019 – 2020 **Blockchain Algorithm Developer (Intern)** HUST Dian Group & HK ASTRI, CN.

Research Publications

* Equal contributions # Corresponding author

Journal Articles

1. G. Guan*, C. Zhao*, **Z. Li***, P. Zhang, Y. Chen, P. Ye, M.-K. Wong, L.-Y. Chan, H. Yan, C. Tang, et al., "EmbSAM: Cell boundary localization and segment anything model for fast images of developing embryos," *Communications Biology*, 2026.
2. G. Guan*, **Z. Li***, Y. Ma*, P. Ye*, J. Cao, M.-K. Wong, V. W. S. Ho, L.-Y. Chan, H. Yan#, C. Tang#, et al., "Cell lineage-resolved embryonic morphological map reveals signaling associated with cell fate and size asymmetry," *Nature Communications*, 2025.
3. **Z. Li#**, Z. Huang, J. Cao#, G. Guan, Z. Zhao, and H. Yan, "An effective method for quantification, visualization, and analysis of 3d cell shape during early embryogenesis," *Quantitative Biology*, 2025.
4. **Z. Li***, D. Xie*, Y. Ma*, C. Zhao, S. You, H. Yan, and Z. Zhao#, "Deep learning-based enhancement of fluorescence labeling for accurate cell lineage tracing during embryogenesis," *Bioinformatics*, 2024.

Conference Proceedings

1. C. Wang*, Z. Huang*, **Z. Li#**, and W. Zhu, "Recover cell tensor: Diffusion-equivalent tensor completion for fluorescence microscopy imaging," in *International Conference on Learning Representations (ICLR)*, 2026.
2. C. Wang, Z. Wang, **Z. Li#**, E. Ozden-Yenigun, and H. Liu, "MSTF-Net: A UAV-Oriented multi-spectral video segmentation method via Modality-Robust, Scale-Adaptive, and consistent fusion," in *The CAAI International Conference on Artificial Intelligence (CICAI)*, 2026.
3. **Z. Li***, C. Wang*, Z. Huang#, Y. Ma, C. Zhao, Z. Zhao, and H. Yan, "Volume tells: Dual cycle-consistent diffusion for 3d fluorescence microscopy de-noising and super-resolution," in *IEEE CVPR*, 2025.

4. **Z. Li**#, Z. Huang, and H. Yan, "3d cellular segmentation of live embryos via topologically and biologically boundary-aware semi-supervised learning," in *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2024.
5. **Z. Li**#, Z. Huang, Z. Zhu, S. You, Z. Zhao, and H. Yan, "Directional and topological transformer with topology priors for 4d cellular image segmentation," in *2024 IEEE International Conference on Image Processing (ICIP)*, 2024.

Preprints and Manuscripts under Review

1. X. Chen, **Z. Li**#, Z. Zhou, H. Chen, C. Wang, and X. Zhu, "Pointing-VLA: Typed spatial grounding interfaces for vision-language-action manipulation," Submitted to *AAAI 2027*, 2026.
2. X. Chen, **Z. Li**, Z. Zhou, X. Zhu, and H. Liu, "Frozen flows forget: Diagnosing and restoring lost motion in a latent-flow world model," *arXiv preprint arXiv:2609.28414*, 2026.
3. **Z. Li***, G. Guan*, X. Xian*, D. Xie, Y. Ma, S. You, Z. Zhu, D. Lee, Z. Zhang, Z. Ran, et al., "CTransformer: Deep-transformer-based 3d cell membrane tracking with subcellular-resolved molecular quantification," Under review at *Nature Methods*, 2026.
4. Z. Zhou*, **Z. Li***#, X. Chen*, Y. Chen, X. Zhu, and H. Liu#, "Adaptive safety filtering for frozen ACC policies via conformal residual calibration," *arXiv preprint arXiv:2609.35415*, 2026.
5. Z. Zhou, **Z. Li**#, X. Chen, J. Li, H. Chen, C. Wang, and X. Zhu, "Think only when needed: Prompt-authority control for selective slow-path intervention in vision-language-action manipulation," Submitted to *AAAI 2027*, 2026.

Honors and Academic Services

Scholarships, Awards and Entrepreneurship

- 2017 National Encouragement Scholarship, HUST, Ministry of Education, China.
- 2020 Outstanding Graduate Honor, HUST.
- Outstanding Graduation Thesis Award, HUST.
- 2025 HK Tech 300 Seed Fund Awardee (PIC), CityUHK.
- 2026 HKSTP Ideation Programme Awardee (PIC), HKSTP.

Academic Services

- 2022 & 2024 Invited Trainer, EMBO Practical Course, European Molecular Biology Laboratory.